

Exploratory Data Analysis for Sales Prediction

Retail Demand Forecasting · Machine Learning

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Data Science · Machine Learning · Business Analytics

Python	Pandas	Seaborn	Matplotlib	Scikit-learn
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9,994	21	4	10
Transactions	Columns	Categories	Key Insights

Introduction to Machine Learning · Module 5 · Final Project

Dataset: Sample Superstore — Kaggle

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1. EXECUTIVE SUMMARY

This portfolio project presents a comprehensive Exploratory Data Analysis (EDA) on a retail sales dataset, carried out as the final project for an Introduction to Machine Learning course. The goal is to identify the key variables that drive sales performance and to build a solid data foundation for training a Sales Prediction model using Linear Regression.

The analysis covers more than 9,994 retail transactions with 21 variables, including product categories, geographic regions, customer segments, shipping modes, discounts, and profit. Through 11 structured analytical sections, the project uncovers 10 actionable business insights that directly inform model design, marketing strategy, and pricing policy.

2. PROJECT OVERVIEW & MOTIVATION

Sales forecasting is a critical capability for any retail or e-commerce business. Accurate demand prediction allows companies to optimize inventory levels, reduce waste, plan marketing campaigns, and improve financial planning. This project tackles that problem using a classic retail dataset and modern data science tools.

Why This Dataset?

- Realistic retail data: The Superstore dataset reflects real-world business complexity with multiple product lines, geographies, and customer types.
- Rich feature set: A mix of numerical (Sales, Profit, Discount, Quantity) and categorical (Category, Region, Segment, Ship Mode) variables enables diverse analytical techniques.
- Business relevance: The insights generated directly apply to supply chain, marketing, and pricing strategy — domains with strong industry demand.
- ML readiness: The dataset structure is ideal for building and evaluating regression models.

3. PROBLEM FORMULATION

Before running any analysis, the team clearly defined the supervised learning problem to ensure every analytical step was aligned with the modeling objective.

Element	Definition	Details
Target Variable (y)	Sales	Continuous monetary value per transaction (USD)
Model Type	Linear Regression	Appropriate for a continuous numeric target variable
Evaluation Metric 1	R ² (R-squared)	Proportion of variance in Sales explained by the model
Evaluation Metric 2	RMSE	Root Mean Square Error — average prediction error in USD

Element	Definition	Details
Feature Variables (X)	Quantity, Profit, Discount, Category, Region, Segment, Sub-Category, Ship Mode	Mix of numerical and categorical variables (categoricals require encoding before modeling)
Expected R ²	0.60 – 0.80	Based on observed correlations in EDA

4. DATASET DESCRIPTION

Source: Kaggle — Sample Superstore Dataset (kaggle.com/datasets/himanshuuike/superstore-sales-dataset)

The Sample Superstore dataset simulates four years of US retail transactions for a fictional superstore, commonly used as a benchmark in business intelligence and data science education. It captures every aspect of a typical retail order lifecycle.

9,994	21	3	4	49
Transactions	Columns	Product Categories	Regions	US States

Column Reference

Column	Type	Description
Row ID	Integer	Unique row identifier
Order ID	String	Unique order identifier
Order Date / Ship Date	Date	Order and shipment date
Ship Mode	Categorical	Standard Class / Second Class / First Class / Same Day
Customer ID / Name	String	Customer identifier and name
Segment	Categorical	Consumer · Corporate · Home Office
Country / City / State	Geo	Geographic identifiers of the transaction
Region	Categorical	West · East · Central · South
Product ID / Name	String	Product identifier and name
Category	Categorical	Furniture · Office Supplies · Technology
Sub-Category	Categorical	17 sub-categories (e.g., Chairs, Phones, Binders...)
Sales	Float — TARGET	Revenue per transaction (USD) — variable to predict
Quantity	Integer	Number of units sold per transaction
Discount	Float [0.0–1.0]	Discount rate applied (0.2 = 20% off)

Column	Type	Description
Profit	Float	Net profit per transaction (can be negative)

5. EDA METHODOLOGY

The exploratory analysis was structured into 11 sequential sections, each designed to answer specific business questions and prepare the dataset for machine learning. The Python libraries used throughout were: **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**.

5.1 Data Loading & Structure

The dataset was loaded using `pd.read_csv()` and immediately inspected with `df.info()`, `df.head()`, and a custom unique-values report. This step confirmed the dataset contains **9,994 rows and 21 columns**. Key findings:

- Data types: 7 object (text), 5 integer, 4 float, 2 date columns.
- Unique value counts revealed that Region has 4 values, Category has 3, and Sub-Category has 17 — confirming the categorical nature of these columns.
- No duplicate rows were detected in the initial scan.

5.2 Descriptive Statistics

Running `df.describe()` produced the following summary for the four numerical variables:

Statistic	Sales (USD)	Profit (USD)	Quantity	Discount
Mean	\$229.86	\$28.66	3.79	0.156
Median	\$54.49	\$8.67	3.00	0.200
Std Dev	\$623.25	\$234.26	2.23	0.157
Min	\$0.44	-\$6,599	1	0.00
Max	\$22,638	\$8,399	14	0.80
Skewness	+3.83	-0.46	+0.78	+0.46

Notable finding: The mean Sales (\$229.86) is more than 4x the median (\$54.49), signaling a heavily right-skewed distribution driven by a small number of very large orders.

5.3 Missing Value Analysis

A systematic null-value audit was performed across all 21 columns (209,874 total cells). **Result: Zero missing values detected.** This is an important finding for ML readiness — no imputation strategy is required before modeling, which eliminates a common source of data leakage and modeling bias.

Null Check Result: 0 missing values across 9,994 rows x 21 columns (209,874 cells).

Implication: Dataset is complete — no imputation required before model training.

5.4 Target Variable Distribution (Sales)

The Sales variable was analyzed using a combined Histogram + KDE plot and a Boxplot. Both visualizations confirmed a strong right-skewed distribution (skewness = +3.83):

- The majority of transactions are low-value (\$0–\$500), with a long tail of high-value orders.
- Mean (\$229.86) significantly exceeds the Median (\$54.49) — confirming positive skew.
- The boxplot revealed numerous outliers above \$2,000, with the maximum reaching \$22,638.

- Modeling implication: A log transformation of Sales (log1p) may be needed to normalize the distribution before training, improving Linear Regression performance.

5.5 Numerical Features Distribution

Histograms and boxplots were generated for Profit, Quantity, and Discount:

Variable	Key Observation	Business Implication
Profit	Contains negative values (min = -\$6,599). Right-skewed distribution.	Some transactions generate losses. Aggressive discounts are the primary cause.
Quantity	Concentrated between 1–5 units. Relatively symmetric distribution.	Predictable purchasing behavior. Low variance simplifies modeling.
Discount	Many transactions at 0 (no discount). Discrete jumps at 10%, 20%, 30%.	Discounts are structured, not random — indicates corporate pricing strategy.

5.6 Correlation Analysis & Heatmap

A Pearson correlation matrix was computed for all numerical variables, visualized as a lower-triangle heatmap using Seaborn's **coolwarm** palette:

Variable Pair	Correlation	Direction	Strength
Profit ↔ Sales	+0.48	Positive	Moderate
Quantity ↔ Sales	+0.20	Positive	Weak
Discount ↔ Profit	-0.22	Negative	Weak-Moderate
Discount ↔ Sales	-0.03	Negative	Very Weak
Quantity ↔ Profit	+0.07	Positive	Very Weak

Insight: Profit is the strongest numerical predictor of Sales. Discount negatively impacts Profit, confirming the discount-destroys-margin hypothesis.

5. EDA METHODOLOGY (continued)

5.7 Category & Sub-Category Analysis

Bar charts and profit-margin tables were generated to compare the three main categories and their 17 sub-categories across Sales volume, Profit, and Profit Margin:

Category	Sales Performance	Profitability	Recommendation
Technology	High sales volume per transaction	HIGHEST margin — strong ROI	Prioritize in marketing & inventory
Office Supplies	High transaction count, moderate volume	Moderate margin, consistent	Reliable revenue base
Furniture	High absolute value per order	LOWEST — sub-categories like Tables generate negative profit	Review pricing & discount strategy

Critical insight: The Tables sub-category within Furniture consistently generates negative profit — selling more Tables actually hurts overall profitability. This counterintuitive finding is only visible through EDA.

5.8 Geographic Analysis (Questions 6 & 7)

Four levels of geographic granularity were analyzed: Region, State, and City.

Geography	Top Performers	Bottom Performers
Region	West (\$725K+) and East (\$678K+) lead	South has lowest total and profit
States (Top 3)	California, New York, Texas	Wyoming, West Virginia, Maine
Cities (Top 3)	New York City, Los Angeles, Seattle	Multiple small cities with <\$200 total sales

5.9 Sales vs. Location Relationship (Question 10)

Boxplots, violin plots, and a Region × Category heatmap were used to assess how strongly geographic location influences sales. **Result: A moderate but significant relationship exists.**

- West and East regions show higher median Sales and average order values.
- High intra-region variability (many outliers) suggests region alone is insufficient as a predictor.
- The Region × Category heatmap revealed interaction effects — e.g., Technology in the West shows significantly higher average sales than any other combination.
- Conclusion: Region should be included as a feature, ideally combined with Category to capture interaction effects.

5.10 Scatter Plot Analysis

Six key variable pairs were visualized using Seaborn's scatterplot with categorical color coding:

- **Profit vs Sales (by Region):** Positive trend; high Sales-high Profit pattern, but with notable dispersion.
- **Quantity vs Sales (by Category):** Weak positive correlation; category differences visible in cluster separation.
- **Discount vs Profit (by Category):** CRITICAL: Clear negative trend. Discounts above 20% systematically erode profit.
- **Discount vs Sales (by Region):** Near-zero correlation — higher discounts do not reliably increase sales volume.
- **Profit vs Quantity (by Category):** Technology cluster shows higher profit per unit; Furniture cluster near zero.
- **Quantity vs Profit (by Segment):** Corporate and Home Office segments show slightly higher profit per unit.

5.11 Discount Impact Analysis

This was one of the most impactful analytical sections. A dedicated visualization plotted average Profit by each discrete discount level (0%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%).

Discounts of 0%–10%: Positive average profit — transactions are profitable.
Discounts of 20%–30%: Profit approaches zero — margin fully consumed.
Discounts $\geq 40\%$: Consistent negative average profit — transactions generate losses.
Business Impact: Discount policy revision could directly improve company profitability without requiring any change to product pricing or sales volume.

5.12 Ship Mode & Customer Segment Analysis

Bar charts and pie charts analyzed the four shipping modes and three customer segments:

- Standard Class is the dominant shipping mode (~60% of all orders) and the highest revenue generator.
 - First Class and Same Day modes have a higher cost structure, reducing their profitability contribution.
 - Consumer is the largest customer segment by transaction count and total revenue.
 - Corporate segment shows a slightly higher average order value, making it a high-value target.
 - Home Office is the smallest segment but shows interest in Technology products.
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6. KEY FINDINGS & BUSINESS INSIGHTS

The EDA surfaced 10 actionable insights with direct implications for both the predictive model design and business strategy.

1 Right-Skewed Sales Distribution HIGH

Mean > 4x Median. Log transformation required before modeling.

2 Technology is Most Profitable Category HIGH

Highest profit margin %. Strategic priority for inventory and marketing.

3 Furniture Generates Sub-Category Losses HIGH

Tables sub-category shows consistent negative profit. Pricing review needed.

4 Discounts >= 20% Destroy Margin CRITICAL

Every discount tier above 20% produces average negative profit.

5 West & East Regions Lead in Sales MED

Together account for the majority of total revenue and transactions.

6 California & New York Are Top States MED

Disproportionate share of sales. Primary marketing targets.

7 Small States Show Minimal Presence LOW

Wyoming, West Virginia, Maine have <\$5K total sales. Expansion opportunity.

8 Region Moderately Influences Sales MED

Significant but not deterministic. Must combine with Category for better signal.

9 Standard Class Dominates Shipping MED

~60% of orders. Optimization of last-mile logistics could reduce costs.

10 Consumer Segment is Largest by Volume MED

Most transactions and highest total revenue. Primary target audience.

Insights Summary Table

#	Insight	Priority	Model Impact
1	Right-Skewed Sales Distribution	HIGH	Feature Selection
2	Technology is Most Profitable Category	HIGH	Feature Selection
3	Furniture Generates Sub-Category Losses	HIGH	Feature Selection
4	Discounts >= 20% Destroy Margin	CRITICAL	Feature Selection
5	West & East Regions Lead in Sales	MED	Supporting Evidence
6	California & New York Are Top States	MED	Supporting Evidence
7	Small States Show Minimal Presence	LOW	Supporting Evidence
8	Region Moderately Influences Sales	MED	Supporting Evidence
9	Standard Class Dominates Shipping	MED	Supporting Evidence
10	Consumer Segment is Largest by Volume	MED	Supporting Evidence

7. DATA CLEANING & PREPROCESSING PLAN

Based on the EDA findings, the following preprocessing steps are required before training the prediction model:

Task	Status	Action Required	Technique
Missing Values	■ Clean	No action needed	No imputation required
Sales Distribution	■ Skewed	Apply log transform	np.log1p(Sales) before modeling
Outliers in Sales	■ Present	Evaluate impact on model	Winsorizing at 99th percentile or log-transform
Categorical Encoding	■ Required	Convert to numeric	One-Hot Encoding (Region, Category, Segment, Ship Mode)
ID Columns	■ Remove	Drop non-predictive columns	Row ID, Order ID, Customer ID, Product ID
Date Columns	Optional	Extract temporal features	Extract Month, Quarter, Year if seasonal analysis needed
Feature Interactions	Optional	Combine features	Create Region_Category composite variable
Feature Scaling	■ Required	Normalize numerical vars	StandardScaler or MinMaxScaler for Quantity, Discount

8. MODEL DESIGN & EXPECTED PERFORMANCE

Baseline Model: Linear Regression
Linear Regression was selected as the initial model because the target variable (Sales) is a continuous numeric value — the classic use case for regression. It also provides interpretable coefficients, making it easy to explain which features have the most impact.

Recommended Feature Set

Feature	Type After Processing	Expected Predictive Power	Notes
Quantity	Numerical	Moderate (+)	Direct volume driver
Profit	Numerical	High (+)	Strongest numerical correlation with Sales
Discount	Numerical	Low (-)	Weak direct effect on Sales
Category	One-Hot (3)	High	Significant differences across categories
Region	One-Hot (4)	Moderate	Geographic demand patterns
Segment	One-Hot (3)	Moderate	Customer type influences order size
Sub-Category	One-Hot (17)	High	Granular product-level patterns
Ship Mode	One-Hot (4)	Low-Moderate	May correlate with order urgency/size

Performance Expectations & Improvement Roadmap

Model Version	Expected R²	Expected RMSE	Approach
Baseline Linear Regression	0.40 – 0.55	~\$400–600	Raw features, no transformations

Model Version	Expected R ²	Expected RMSE	Approach
Improved Linear Regression	0.60 – 0.75	~\$200–350	Log(Sales), One-Hot encoding, feature scaling
Random Forest Regressor	0.75 – 0.85	~\$150–250	Captures non-linear relationships & interactions
Gradient Boosting (XGBoost)	0.82 – 0.90	~\$100–200	Best performance for tabular data; handles outliers well

9. TECHNICAL SKILLS DEMONSTRATED

This project showcases a comprehensive set of data science competencies across the full EDA workflow — from raw data ingestion to model design readiness.

Skill Area	Techniques Used	Libraries/Tools
Data Ingestion & Inspection	CSV loading, shape analysis, dtype review, unique value profiling, null detection	Pandas (read_csv, info, describe, isnull)
Statistical Analysis	Descriptive stats, skewness, percentiles, correlation matrix (Pearson), mean vs median	Pandas, NumPy
Data Visualization	Histograms, KDE plots, Boxplots, Violin plots, Bar charts, Heatmaps, Scatter plots, Pie charts	Matplotlib, Seaborn
Business Analysis	Profitability by category, geographic segmentation, discount impact, customer segmentation	Pandas GroupBy, Pivot Tables
Feature Engineering (Planned)	Log transformation, One-Hot encoding, interaction terms, date feature extraction	Pandas, Scikit-learn
ML Model Design	Target/feature definition, problem framing, metric selection (R^2 , RMSE)	Scikit-learn (planned)
Code Quality	Modular code, consistent naming conventions, commented sections, visual configuration	Python best practices
Communication	Bilingual documentation (Spanish/English), clear chart labeling, structured insights	Jupyter Notebook, Markdown

10. CONCLUSIONS & NEXT STEPS

This exploratory data analysis successfully accomplished its primary objective: understanding the structure, distributions, correlations, and business patterns within the Superstore retail dataset — and translating those findings into a concrete ML modeling plan.

What Was Achieved

- Confirmed dataset completeness: zero missing values across 209,874 cells.
- Identified the target variable distribution and its preprocessing requirements.
- Determined the most predictive features based on statistical correlation and business logic.
- Answered 10 specific business questions through targeted visualizations.
- Built a clear data pipeline and modeling roadmap ready for the next project phase.

Next Steps — Phase 2: Model Training

#	Step	Description
1	Data Preprocessing	Apply log transformation to Sales, One-Hot encode categoricals, remove ID columns, scale numerical features.

#	Step	Description
2	Train/Test Split	Split dataset 80/20 with stratification by Region to ensure representative samples.
3	Baseline Model	Train Linear Regression with sklearn. Evaluate R ² and RMSE on test set.
4	Feature Importance	Analyze regression coefficients to validate EDA feature selection hypotheses.
5	Model Improvement	Test Random Forest and XGBoost regressors. Apply cross-validation.
6	Hyperparameter Tuning	Use GridSearchCV or RandomizedSearchCV to optimize model parameters.
7	Final Evaluation	Compare all models on test set. Select best performer. Document results.
8	Deployment (Optional)	Wrap model in a simple prediction API using FastAPI or Streamlit dashboard.

This project represents the first step of a full Machine Learning pipeline, demonstrating end-to-end analytical thinking from raw data to actionable business insights.

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